

VASUNDHARA S R

Thiruvananthapuram, Kerala, India | +91-9539010497 | vasundharasr.331@gmail.com | linkedin.com/in/vasundhara-s-r | github.com/Vasundhara-331

PROFESSIONAL SUMMARY

Final-year Computer Science Engineering student with strong fundamentals in Object-Oriented Programming (Python, Java), SQL/RDBMS design, and RESTful API and microservices architecture, built across 3 industry internships. Hands-on experience with Generative AI foundations, Docker, and security-conscious engineering: Role-Based Access Control, DevSecOps workflows, and multi-tenant data isolation in production systems. Two-time Smart India Hackathon qualifier and open-source contributor (4+ community forks on a published repository).

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C, HTML5, CSS3

Frameworks & Databases: FastAPI, Flask, React.js, Node.js, PostgreSQL, MySQL, MongoDB

Cloud, DevOps & Security: Docker, Git/GitHub, REST API Design, RBAC, DevSecOps, Multi-Tenant Data Isolation, Agile Frameworks

AI & Agentic Engineering: Agentic AI, LLM Gateways, Graph-Based Code Analysis (Neo4j/Cypher), Prompt Engineering, NLP

INTERNSHIP EXPERIENCE

Artificial Intelligence Intern | IBS Software - Trivandrum, Kerala

Jun 2026 - Jul 2026

- Built REST APIs (FastAPI) integrating a LiteLLM-based gateway that unifies routing, authentication, and token management across multiple LLM providers, with Server-Sent Events for real-time response streaming.
- Evaluated **3 graph databases** (FalkorDB, KuzuDB, LadybugDB) and migrated to Neo4j for stability, then indexed a 1000+ file enterprise repository into queryable code graphs using Cypher to support AI-assisted code understanding.
- Ran a structured comparative study of traditional debugging vs. two graph-context-assisted methods across Jira-tracked bugs, benchmarking accuracy, latency, and token cost - findings directly informed the team's tool-selection guidance for large-repository maintenance.

Full-Stack Web Development Intern | MBCET (In-House) - Trivandrum, Kerala

Jul 2025 - Jan 2026

- Led requirement gathering with academic stakeholders and architected **2 production-ready platforms** - a department information portal and an alumni networking ecosystem using React.js, TypeScript, and Cloudinary.

Full-Stack Web Developer | Triassic Solutions Pvt. Ltd. - Technopark, Trivandrum

Jun 2025

- Co-engineered MaxProfit, a multi-organization financial intelligence platform (Flask, React.js, MySQL), designing high-throughput schemas and interactive charts for low-latency profitability reporting.

Research Intern | Indian Institute of Space Science and Technology (IIST) - Trivandrum, Kerala

Dec 2024 - Jan 2025

- Evaluated post-quantum-resilient cryptographic architectures for IoT and blockchain applications.

KEY PROJECTS

Novara: Self-Learning Middleware for Token Optimization (Capstone) - Python, FastAPI, Docker, PostgreSQL, LiteLLM

- Architecting a middleware gateway that routes requests across multiple LLM providers (OpenAI, Claude, Gemini) to optimize cost, response quality, and governance for enterprise applications.
- Designing a self-learning optimizer to auto-tune prompt compression, retrieval depth, and model selection.

ZYRA: AI-Driven Web Development Agent - Ollama LLM (Phi-3), TypeScript, React, Node.js, Firebase

- Built the SRS Generation module (offline local LLM inference) converting natural-language prompts into requirement docs with multi-format export, cutting external API costs to **\$0**.

Sentry: Document Automation for Kochi Metro Rail Limited - React.js, TypeScript, Supabase, Tesseract OCR, NLP · SIH 2025 Qualifier

- Built an AI-powered document automation platform with OCR extraction, NLP-driven auto-classification, and semantic search, RBAC and admin analytics - qualified nationally at Smart India Hackathon 2025.

MaxProfit: Financial Metrics Tracking Platform - React.js, Flask, Python, MySQL, Docker · 4+ GitHub forks

- Built a full-stack financial intelligence platform with time-series forecasting and multi-organization comparison dashboards over a high-throughput MySQL schema.
- Published as a containerized, modular open-source repository adopted with **4+ community forks** since release.

Built and iterated using AI-assisted development tools: Cursor, GitHub Copilot, Claude, Windsurf

EDUCATION

B.Tech in Computer Science and Engineering - Mar Baselios College of Engineering and Technology (CGPA: 8.98/10)

2023 - 2027

Higher Secondary Education - Carmel Higher Secondary School (98.3%)

2022 - 2023

Secondary Education - Holy Angels' ISC School (97.8%)

2020 - 2021

CERTIFICATIONS & ACHIEVEMENTS

Certifications: AI Agent Architect (IBM SkillsBuild) · AWS Academy - Generative AI Foundations

Achievements: Smart India Hackathon Qualifier (2024 and 2025) · #10 Position in Top 12 Karma Miners by GTech MuLearn

Leadership: Chief Vibe Officer, Catalyst IEDC MBCET (2026 - Present); previously Chief Executive Officer (2025 - 26)

Editorial Board Member, College Magazine (2025 - 26); Vice Chairperson, Adyatva Science Association, MBCET (2024 - 25); Co-Lead - Creative Wing, CSI SB MBCET (2024 - 25)